

MediTrack: Readmission Risk Prediction & Decision Support

Point-of-Discharge Decision Support System Grounded in Published Clinical Guidelines

Project Brief: CP-02 | Healthcare Analytics & Clinical AI

Cohort: CSE VII | Batch 2, Room 205, Block 3 | Sharda University

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Core Capabilities:

- 30-Day Hospital Readmission Scoring at Point of Discharge with Stated Decision Threshold
- Patient-Level Transparent Factor Attribution using SHAP Tree Explainability
- Guideline-Grounded Assistant (ADA Standards, AHRQ Project RED, CMS TCM) with Mandatory Source Citations
- Strict Safety Guardrails Defending Against Diagnostic and Prescriptive Violations
- Hospital-Wide Clinical Quality Dashboard with High-Risk Cohort Drill-Through

Clinical Problem Statement & Regulatory Context

The Challenge of Post-Discharge Readmission and Need for Real-Time Decision Support

The Healthcare Burden:

- Hospital readmission within 30 days of discharge is a critical quality benchmark tracked under CMS HRRP.
- Avoidable readmissions account for billions in excess expenditure, hospital financial penalties, and patient morbidity.
- **The Current Gap:** Most healthcare systems analyze readmissions *retrospectively* via post-facto billing audits.

The MediTrack Solution:

- Intervene *proactively at the point of discharge* before the patient departs the hospital.
- Combine statistical risk scoring with individualized clinical factor transparency.
- Provide the care team with an intelligent transitional care assistant grounded in national standards.

Dataset Architecture & Data Pipeline

UCI Diabetes 130-US Hospitals (1999–2008) Data Quality & Preprocessing

Primary Ingestion & Clinical Exclusions:

- Initial encounters loaded: **101,766** records across 10 years of multi-hospital clinical care.
- **Documented Exclusion Rules Applied:** Excluded **2,423** deceased and hospice discharges (IDs 11, 13, 14, 19, 20, 21).
- **Post-Exclusion Cohort:** **99,340** validated encounters and **69,987** unique patients.

Feature Engineering & High-Cardinality Groupings:

- **Missing Fields:** Weight (>96% missing) mapped to recording status flag; Payer Code (~40% missing) categorized.
- **ICD-9 Mapping:** Over 800 diagnosis codes consolidated into 9 standard clinical organ categories.
- **Medication Harmonization:** 23 diabetes medications tracked for active use, dosage titration (Up/Down), and polypharmacy.

Exploratory Findings & Statistical Hypothesis Testing

Rigorous Statistical Basis for Core Clinical Risk Determinants

1. Base Readmission Rate:

- 30-Day Readmission (<30 days): **11,314** encounters = **11.39%** (95% CI: 11.19% – 11.59%).

2. Statistical Hypothesis Tests:

- **Age Group:** Pearson's Chi-Square $\chi^2 = 133.625$, $df = 9$, $p = 2.13 \times 10^{-24}$ (Statistically Significant).
- **Admission Type:** $\chi^2 = 32.360$, $df = 6$, $p = 1.39 \times 10^{-5}$. Emergency vs Elective Odds Ratio = **1.145**.
- **Prior Inpatient History:** Mann-Whitney U $p = 0.000$, Welch's t $p = 4.87 \times 10^{-264}$.
 - Readmitted patients mean prior stays = **1.223** vs Non-readmitted = **0.555**.
 - Logistic Odds Ratio = **1.337** per additional prior visit (95% CI: [1.321, 1.353]).

Relational Database & Analytical Queries

Normalized SQLite Relational Architecture & Runnable SQL Workflows

Relational Schema (4 Normalized Entities):

- **patients:** 69,987 unique patients (demographics, longitudinal count).
- **encounters:** 99,340 inpatient stays (admission type, LOS, labs, utilization, readmission flag).
- **diagnoses:** 298,020 records (primary, secondary, tertiary ICD-9 and clinical classifications).
- **medications:** 117,953 active diabetic prescription entries across 23 tracked therapies.

Executed Analytical Queries:

- **Query 1 (Cohort Readmission):** Patients with 3+ prior visits show readmission rates exceeding **44.8%**.
- **Query 2 (Stay by Diagnosis):** Neoplasms leads length of stay (5.28 days); Diabetes leads readmission rate (13.10%).
- **Query 3 (Encounter Sequences):** Window functions (ROW_NUMBER, LAG) trace consecutive admissions per patient.

Predictive Modeling & Class Imbalance Strategy

Benchmark Evaluation of Linear, Tree Ensembles, and Neural Baseline

Model Performance Benchmark (Holdout Test: 19,868 Encounters):

- **Logistic Regression (Balanced):** ROC-AUC: **0.6383** | PR-AUC: **0.2006** | Brier: 0.2319
- **Random Forest (Balanced):** ROC-AUC: **0.6447** | PR-AUC: **0.2064** | Brier: 0.2147
- **Gradient Boosting (Balanced) [BEST]:** ROC-AUC: **0.6445** | PR-AUC: **0.2064** | Brier: 0.2266
- **Neural Baseline (MLP):** ROC-AUC: **0.6421** | PR-AUC: **0.2059** | Brier: 0.0976

Class Imbalance Strategy Justification:

- Applied cost-sensitive class weighting ($w_1 \approx 4.39$).
- Justification: Preserves authentic clinical feature covariance and avoids synthetic artifact generation in discrete ICD-9/medication vectors.

Clinical Decision Threshold Justification

Defending the Recall vs. Precision Trade-off in Clinical Practice

Why Naive 0.50 Threshold Fails Clinically:

- In the unweighted neural baseline, a 0.50 threshold yields **<1% Sensitivity** (misses 99% of readmissions!).
- Default probabilistic thresholds are calibrated to base rates, not clinical utility.

Tuned Operational Decision Threshold (0.18):

- **Clinical Sensitivity (Recall): 65.5%** — Successfully flags ~2 out of every 3 preventable readmissions.
- **Cost-Benefit Asymmetry Defense:**
 - **Cost of False Negative:** ~\$15,200 (avoidable ICU/ER admission, CMS penalty, clinical decompensation).
 - **Cost of False Positive:** ~\$150 (transitional nurse phone outreach, medication review, primary care check).
 - The 1:100 cost ratio heavily favors high sensitivity over specificity in transitional care triage.

Text Processing & Patient-Level Factor Transparency

ICD-9 Diagnosis NLP Extraction & SHAP Factor Attribution

Text Processing & NLP Feature Derivation:

- Derived structured clinical descriptions from ICD-9 codes (e.g. congestive heart failure, acute MI, CKD).
- Extracted 50 TF-IDF n-gram clinical term features and evaluated incremental performance gain.
- Finding: Reinforced organ-specific decompensations without overfitting.

SHAP Factor-Level Explainability:

- Generates individualized patient feature contributions (waterfall attributions) for every encounter.
- Clinicians can see exactly why a patient was flagged (e.g., prior inpatient visits: +8.4%, insulin titration: +4.2%).
- Eliminates black-box distrust and guides targeted discharge actions.

Guideline Retrieval Layer & Clinical Safety Guardrails

Grounded AI Decision Support with Mandatory Citations & Refusal Protocols

Published Clinical Knowledge Base:

- **ADA Standards of Care (2024):** Glycemic discharge planning, insulin education, Rule of 15 hypoglycemia.
- **AHRQ Project RED:** 12-component re-engineered discharge protocol, teach-back method, AHOP plan.
- **CMS TCM Services:** 48-hour interactive outreach and 7-day primary physician follow-up.
- **Mandatory Citations:** Every recommendation includes full publication title, guideline section, and evidence grade.

Clinical Safety Guardrails:

- **Refusal 1:** Diagnostic queries ('Do I have cancer?', 'Diagnose illness') are strictly blocked.
- **Refusal 2:** Prescriptive requests ('Prescribe 500mg Metformin', 'Change dosage') are immediately refused.
- The assistant defers all prescriptive and diagnostic authority to the licensed attending medical team.

Quality Dashboard, Verification & Limitations

Healthcare Economics, Fairness Parity, Drift Monitoring & Clinical Limitations

Verified System Deliverables:

- **Interactive Web Application:** Encounter selector, real-time scoring, SHAP waterfall, guardrailed chat, triage worklist.
- **Financial ROI Simulator:** Net **\$3.08M** hospital savings per 10,000 discharges (ROI: 154.8%).
- **Fairness & Drift Audits:** Verified parity across race/gender (EEOC 4/5 rule satisfied) and healthy feature stability (PSI < 0.05).

Documented Clinical Limitations:

- Dataset lacks outpatient lab values (e.g. serial eGFR, Troponin) and social determinants of health (SDOH).
- Historical data (1999–2008) predates contemporary SGLT2i / GLP-1 RA medication guidelines.
- Prospective validation in live clinical EHR workflow is required prior to autonomous clinical deployment.